

NPTEL

Long Short-Term Memory

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Long Short-Term Memory (LSTM)

“My grandparents live in **Germany** and I visit them every summer, which is why I understand **German** quite well”

The connection between the location and the language is separated by many words.

Long term dependency in sequential data

Recurrent Neural Network (RNN) cannot capture this long-term dependency in sequential data due to vanishing gradient problem

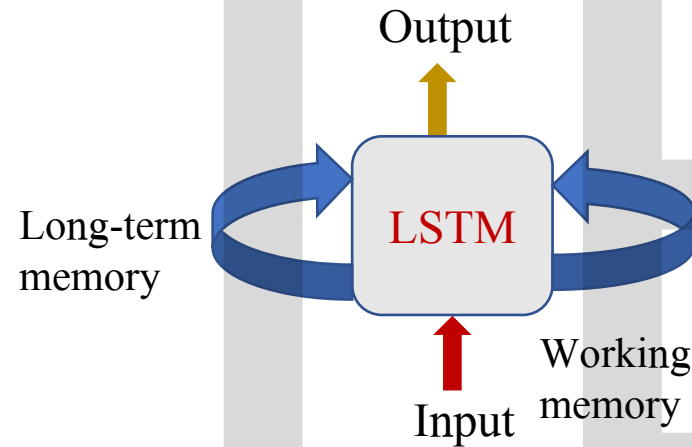
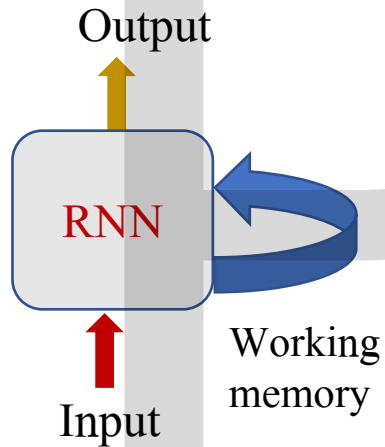
Ref:

Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.

Difference between RNN and LSTM

RNNs are designed to handle sequential data by maintaining a hidden state that captures information from previous time steps.

But they can't handle long term dependencies



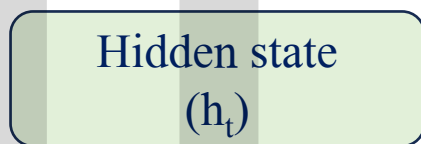
LSTMs introduce a memory cell that holds information over extended periods.

This addresses the challenge of learning long-term dependencies.

What are states in an LSTM

States are the internal memories that carry information across time steps.

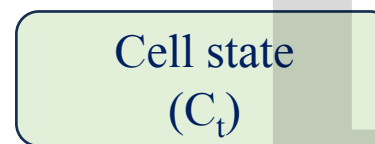
There are **two main states in LSTM**:



↓
Short term memory

Used for predictions and passed to the next time step.

This was already there in vanilla RNN

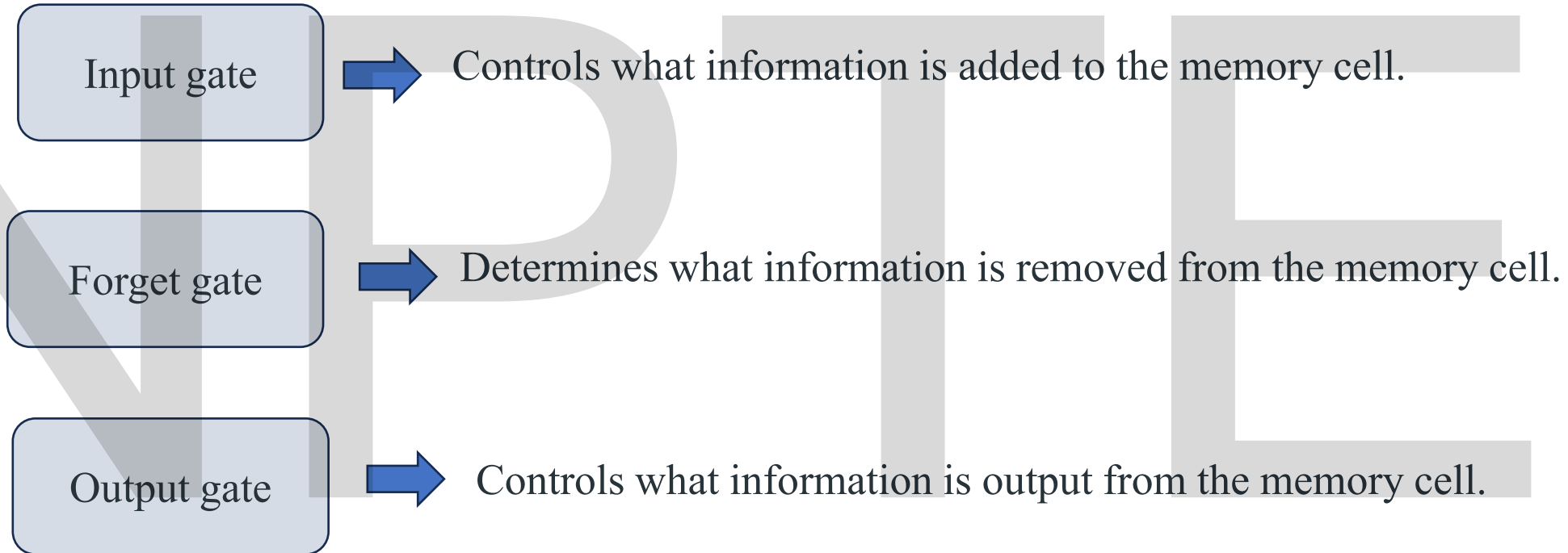


↓
Long term memory

Carries important information across many time steps

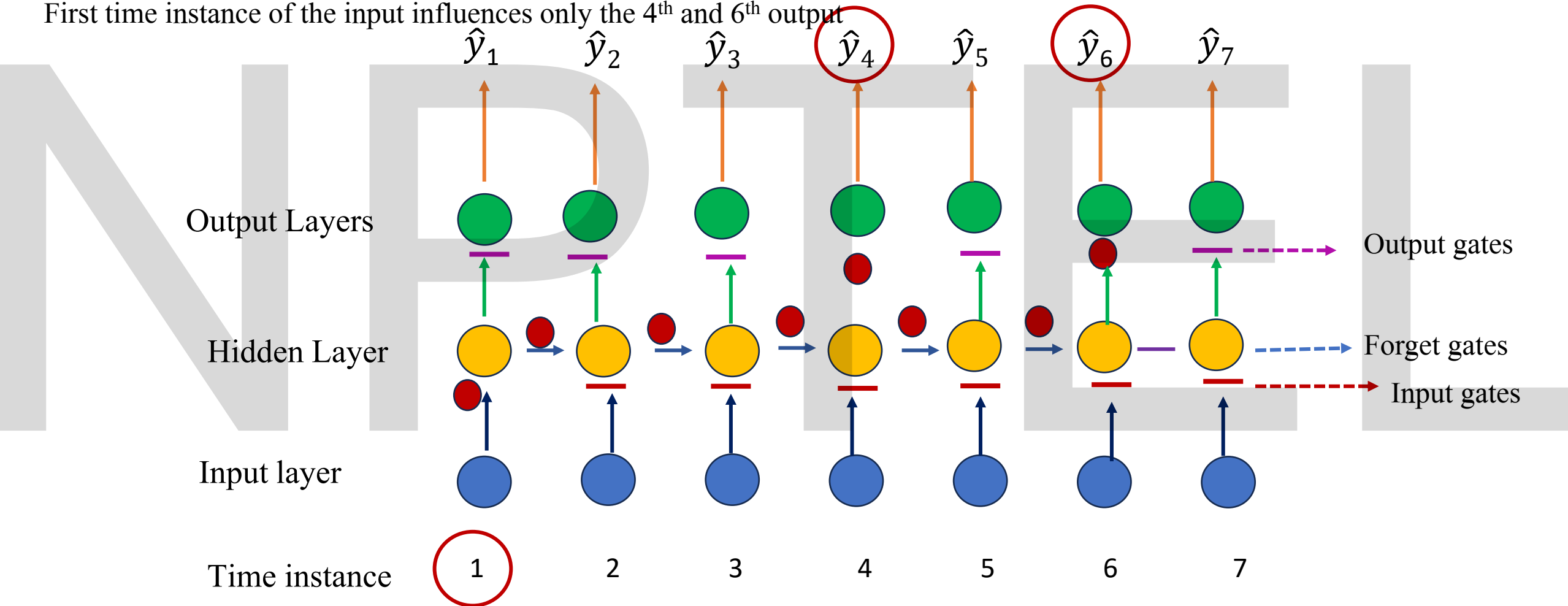
LSTM Architecture

It has three gates

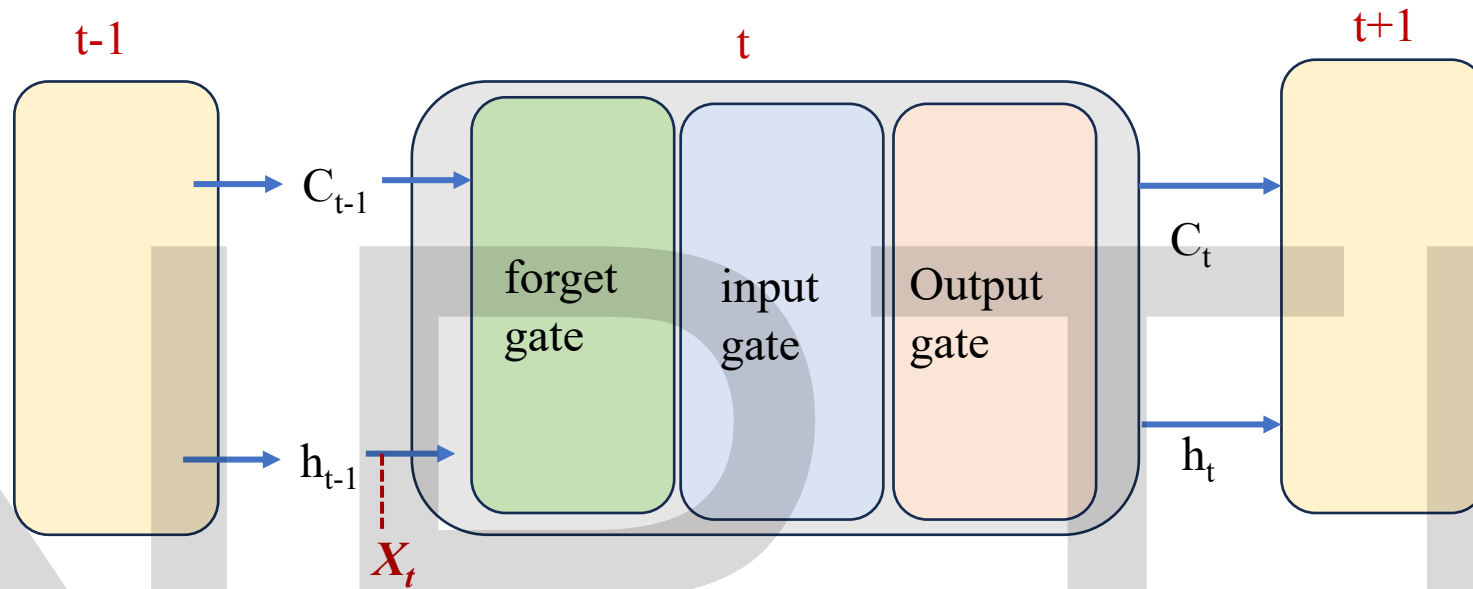


Ex: Allowed students may attend and participate today.

First time instance of the input influences only the 4th and 6th output



LSTM Architecture



LSTM Chain with three nodes

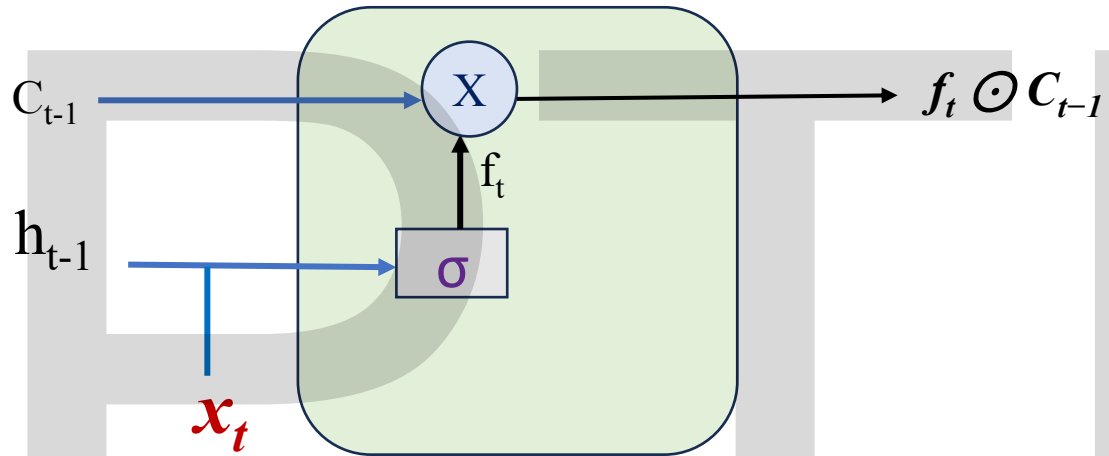
C_{t-1} : Cell state from the previous time step. (Long term memory that carries information through the sequence)

h_{t-1} : Hidden state from the previous time step. (short term memory)

X_t : Current input vector at time step t .

- ✓ Gates are constructed in such a way that they can be closed or opened at any instance of time, which makes them flexible.
- ✓ Gates are also neural network with learnable functions

Architecture of Forget Gate



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

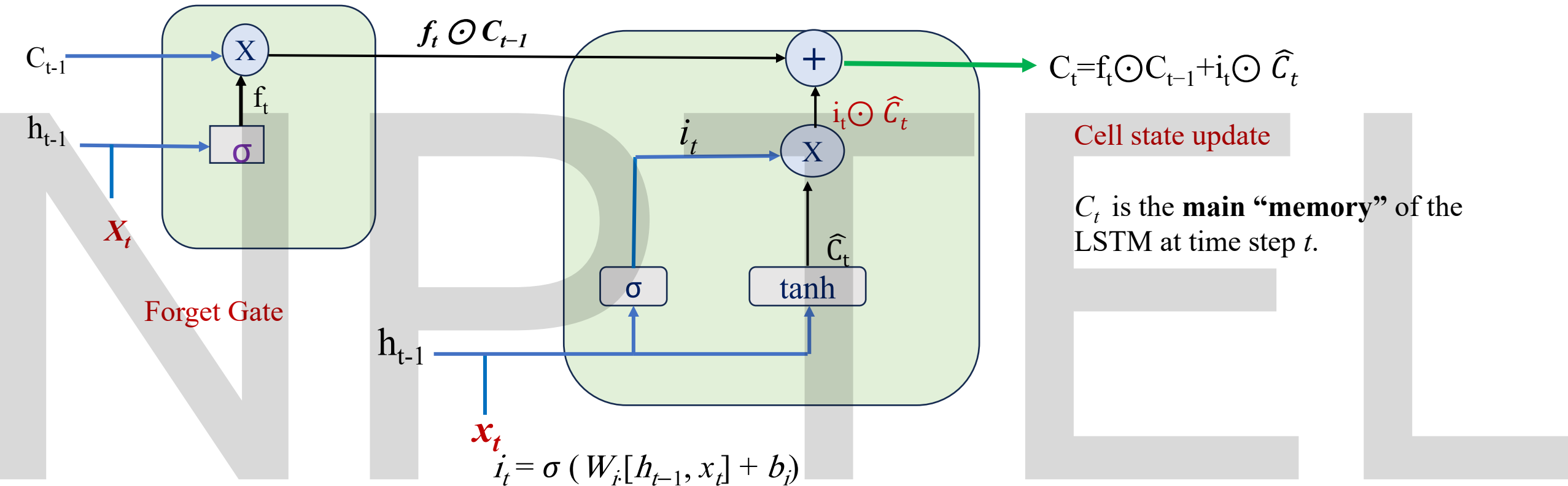
W_f : weight matrix associated with the forget gate

Value of f_t lies between 0 and 1

0: Completely forget

1: Completely remember

Architecture of Input Gate



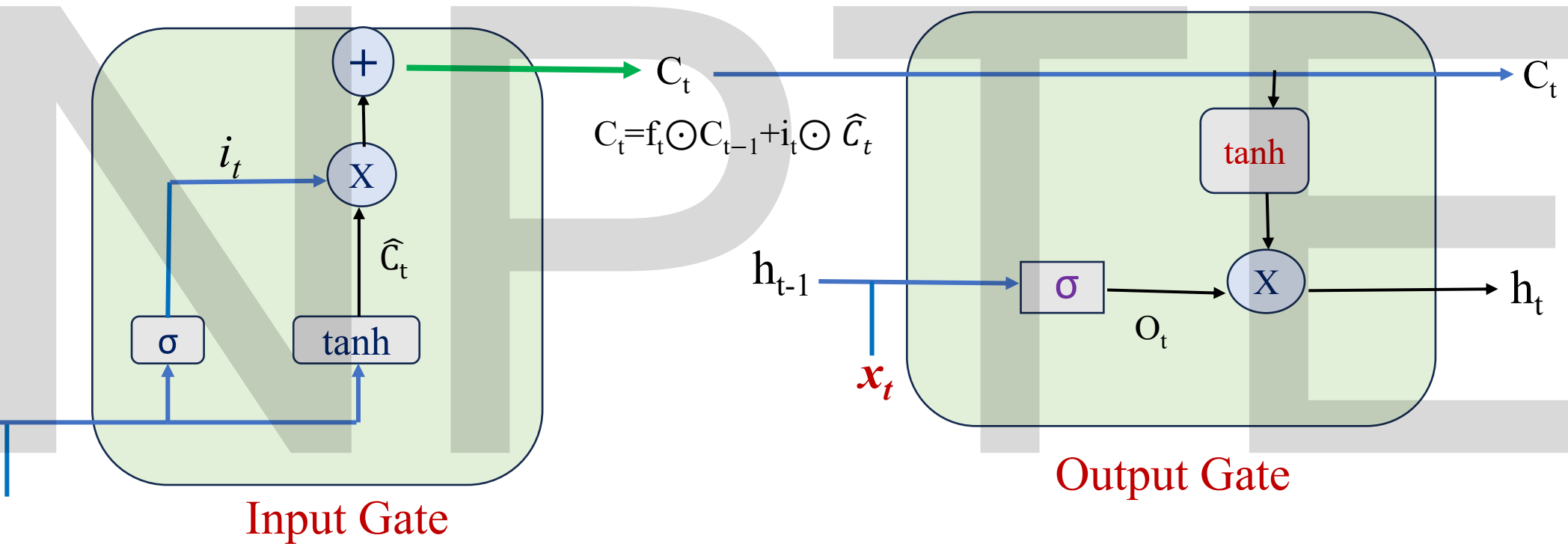
$$\hat{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

W_i : weight matrix associated with the input gate

W_c : weight matrix associated with the candidate cell

Candidate cell is the new information that could be added to the cell state at the current time step.

Architecture of Output Gate



$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

W_o : weight matrix associated with the output gate

$$h_t = o_t \odot \tanh(C_t)$$

How Vanishing Gradient Problem of RNN is mitigated in LSTM

Cell state update in LSTM is given by

$$C_t = f_t \odot C_{t-1} + i_t \odot \hat{C}_t$$


When $f_t \approx 1$, then

$$C_t \approx C_{t-1} + i_t \odot \hat{C}_t$$

Gradient can flow in the backward direction

When $f_t \approx 0$, then

$$C_t \approx i_t \odot \hat{C}_t$$

Due to the presence of the additive term, gradient doesn't vanish

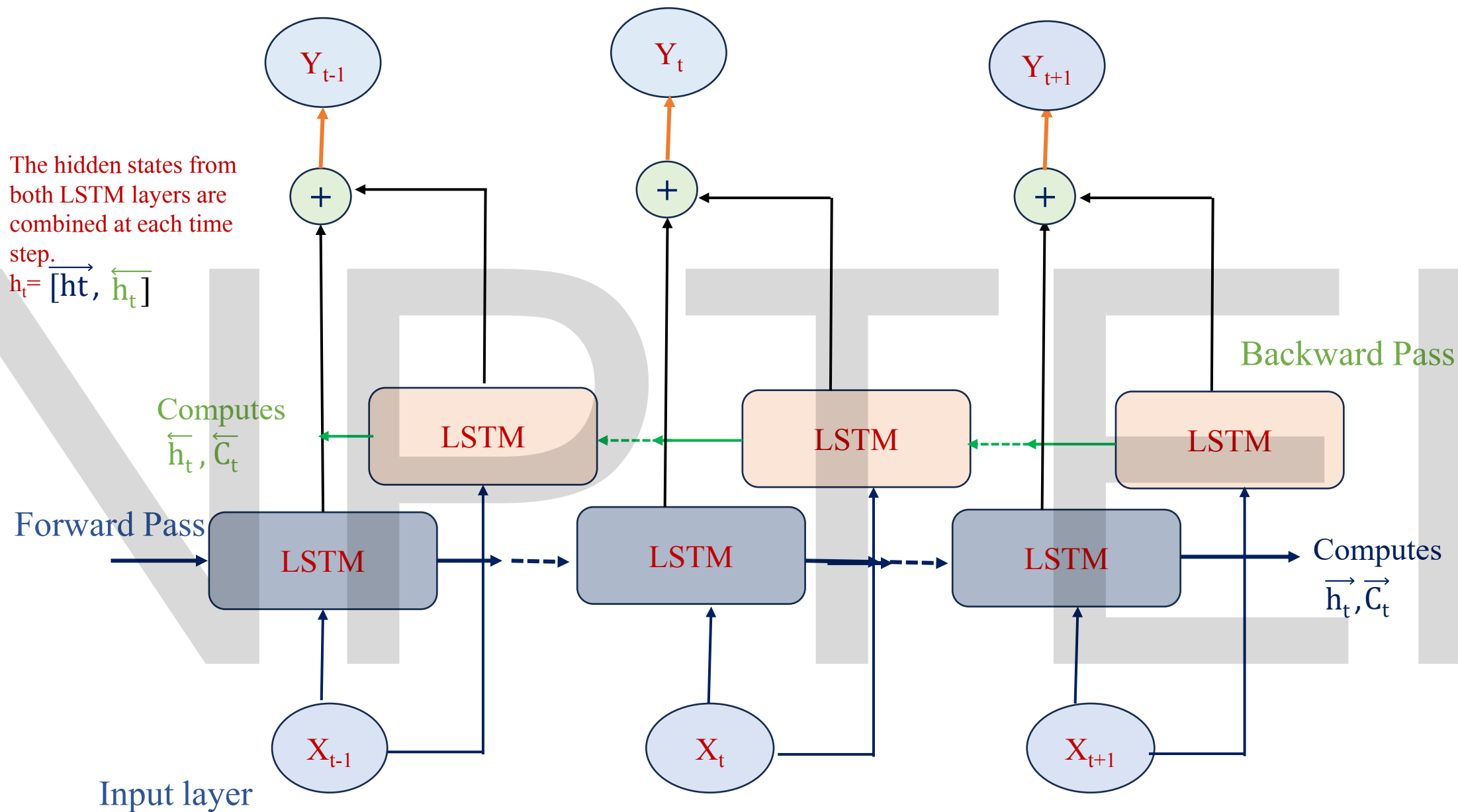
Bidirectional LSTM

Ex: “I can **book** a flight.”

We want to do parts of speech tagging for the word **book**

- Bi-LSTM is a type of recurrent neural network (RNN) that processes sequential data in both forward and backward directions.
- This allows the model to capture both past and future context of the input sequence.

The hidden states from both LSTM layers are combined at each time step.
 $\vec{h}_t = [\vec{h}_t, \overleftarrow{h}_t]$



Bidirectional LSTM

Stacked LSTM

In stacked LSTM, (Deep LSTM) multiple LSTM layers are placed one on top of another.

Example text:

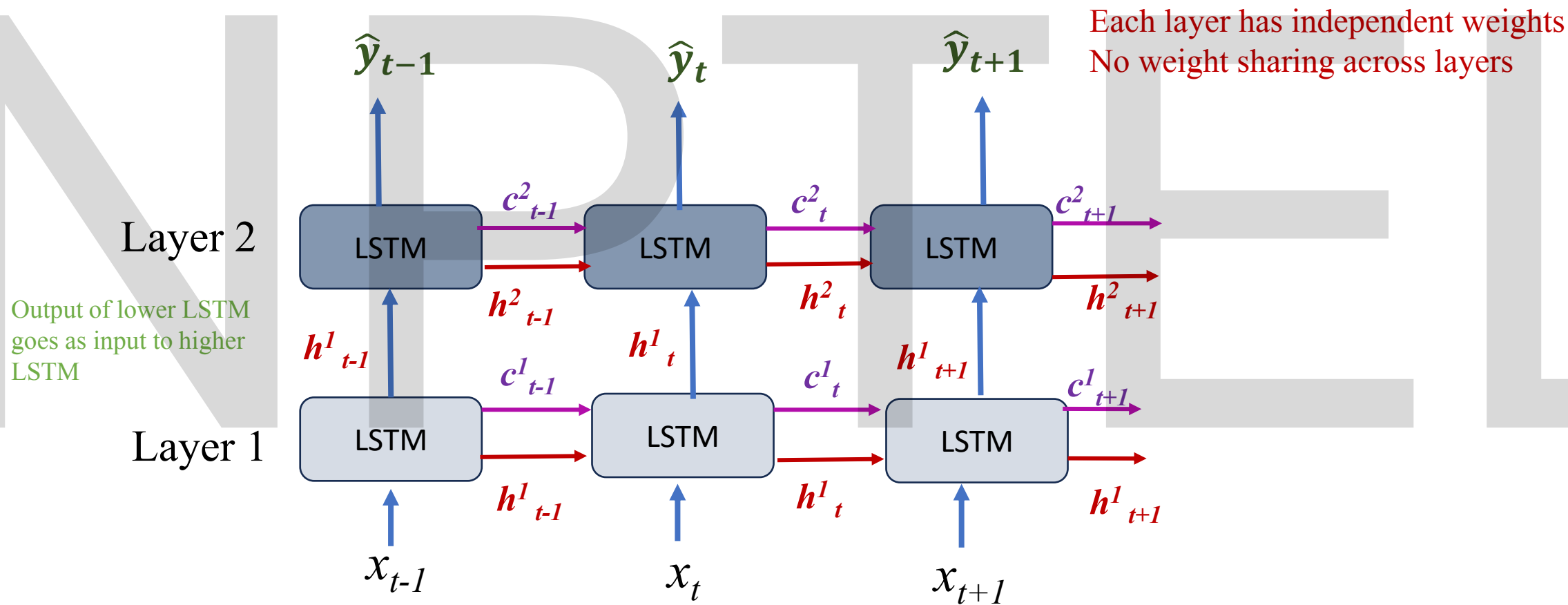
“I grew up in France and I speak French”

Layer 1: learns grammar, word dependencies

Layer 2: learns semantic relationship (France → French)

A single-layer LSTM may fail to capture complex hierarchical features

Stacked LSTM



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Summary

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Next Session
Gated Recurrent Unit

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Thank you